Higher explanation-necessity scores are associated with longer written explanations of autonomous driving scenes

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ABSTRACT

Explanations of automated-vehicle behavior may help occupants understand what prompted a maneuver, but how much an explanation should say remains an open design question. Retrospective annotation of recorded driving scenes can supply candidate content. We analyzed 1,103 training and test records from the release, matching the source-described corpus size; each paired human-written text with an aggregate perceived explanation-necessity score. Explanations contained 9,541 whitespace-delimited tokens (mean 8.65; range 2–20) before lexical preprocessing. Higher scores were associated with longer explanations (Pearson's $r = 0.374$; template-cluster bootstrap 95% interval [0.266, 0.448]), including after score-1.0 observations were excluded. We call this corpus-level length association *progressive elaboration*. Bounded-model comparisons did not identify a unique latent-category count. For groupwise summaries, we used four descriptive partitions: three below-ceiling regions and 59 records scored 1.0. Record-weighted and unique-template analyses showed higher within-partition lexical distinctiveness in upper than in lower partitions. In template-grouped cross-validation of these exploratory partitions, the unweighted text model exceeded token-count and stratified-dummy baselines in balanced accuracy (0.382 versus 0.327 and 0.247). The 1,280-record rerun preserved the score–length, lexical, and predictive findings and left the latent-category count unresolved. These results motivate prospective studies of how annotation-derived content should be selected, compressed, and delivered in real time.

Introduction

When an automated vehicle slows, stops, or changes course, an occupant may observe the maneuver without knowing what prompted it. Explanations of automated-vehicle behavior have therefore been studied in relation to understanding, self-reported trust, and situation awareness[1–5]. Reported effects are context-dependent, varying with explanation content and timing, perceived risk, modality, and information load[2,4–6]. The design question therefore concerns not only *what* to explain, but also *when* an explanation is wanted and how much content it should contain[7,8].

Retrospective annotation and listening during travel provide different perspectives on explanation content. A passenger may request or process an explanation while monitoring a moving vehicle. An observer who retrospectively annotates a recorded scene acts as a speaker–observer, articulating scene-specific details without that concurrent demand. For listener-side delivery, a simulator study of fully automated driving found that the effects of multimodal explanations differed depending on whether passengers were engaged in a non-driving task[9]. While prospective evaluation is needed to determine which details, in what form and amount, should reach a passenger, such evaluation needs candidate content to work with. Retrospective annotation is one way to obtain it, providing scene-specific material that can then be selected, compressed, or staged for delivery.

Two distinct datasets illustrate these settings. TimelyTale[10,11] records in-situ passenger requests together with multimodal driving and passenger context to study when explanation demand arises during highly automated travel. Shen et al.[8] introduced a retrospective driving-video dataset containing first-person explanations and bounded human judgments of explanation necessity. We analyze the current release of Shen et al.'s dataset, using its 1,103 training and test records as the primary cohort and all 1,280 released records as a sensitivity analysis of cohort scope. Each analyzed row contains one aggregate bounded necessity score and one human-written first-person explanation. A scene judged to need an explanation could be described briefly or at greater length. For listener studies that draw messages from this corpus, it matters whether selecting across necessity levels also selects texts of different lengths. We therefore ask whether records with higher aggregate necessity scores contain longer human-written explanations.

The necessity score and explanation length require explicit operational definitions. The source dataset termed the score "criticality," but the public schema does not operationalize objective risk: it contains no fields for collision probability, time-to-collision, hazard severity, workload, physiology, or driving performance. We therefore use *perceived explanation necessity* for

the construct and reserve "criticality" for legacy field names. Explanation length is operationalized as whitespace-delimited token count before lexical preprocessing; the 1,103 human-written explanations contain 9,541 such tokens in total.

A further question is whether the necessity scores form distinct groups or vary along a continuum. Identifying distinct categories would require evidence that the score distribution supports a particular number of latent groups. We use a Gaussian mixture model (GMM) to summarize the distribution with overlapping components and obtain a reference assignment without prespecifying score cutoffs. Adding components to a Gaussian mixture can improve its approximation of a skewed or otherwise non-normal continuous distribution without revealing distinct categories[12]. We therefore compare bounded continuous and mixture models by how well they predict scores held out from fitting[13,14].

In this paper, we address three questions using this retrospective annotation corpus: (1) how bounded continuous and mixture models compare out of sample and whether that comparison supports a unique latent-category count; (2) how human-written explanation length varies with perceived explanation necessity; and (3) how exploratory score partitions differ in corpus-specific lexical distributions under record-weighted and unique-template estimands. For the third question, we summarize lexical distinctiveness under two ways of weighting repeated wording. A normalized-message template groups messages that become identical after case and spacing are standardized and specified punctuation is removed. The record-weighted analysis counts every occurrence, whereas the unique-template analysis counts each template once in every partition where it appears. The first summary reflects lexical overlap at the observed message frequencies; the second reflects overlap among distinct normalized wordings within each partition. A concrete example is given below. We additionally audit preprocessing, negation, repeated text templates, and leakage in a predictive text analysis. Partition-level analyses are descriptive, and maximum-score observations are analyzed as an annotation-ceiling category.

Results

The main finding was a positive association between aggregate necessity scores and annotation length. We first report the data-audit and score-distribution findings, then examine this association, lexical profiles, and text prediction.

Release-audit findings relevant to interpretation

The release audit found no complete-row duplicates, but it revealed two features that affect interpretation: the absence of writer-level links and substantial reuse of wording. Dataset provenance, cohort selection, and processing procedures are described in Methods; the full identifier audit is in Supplementary Table S1.

Each released record pairs an aggregate necessity score with one retained explanation. The release does not identify the writer or link that explanation to the writer's own necessity rating. It also lacks the individual ratings and candidate explanations from which the released record was assembled. The score–length association therefore describes records in the corpus, not how individual writers changed their explanations as their own ratings changed.

The 1,103 records contained 594 normalized-message templates; 600 records belonged to one of 91 templates that appeared more than once. For illustration, "I am slowing down." and "i am slowing down!" share the template `i am slowing down`. This is a constructed example of normalization, not a quotation from the dataset. Templates group matching wording, not paraphrases or explanations with similar meanings. We used these groups to assess how repeated wording affected uncertainty estimates and prediction on unseen templates.

Bounded score distribution and exploratory reference partitions

The score distribution contained repeated values and concentrations at its endpoints and midpoint; 78.5% of records had a score shared by another record. Twenty-one records were exactly 0, 17 were exactly 0.5, and 59 were exactly 1. The ceiling category thus comprised 5.35% of the 1,103 records (exact binomial 95% CI [4.10%, 6.85%]). Almost all scores (1,102 of 1,103) lay close to a regular numerical grid with spacing $1/1836$. These features can influence the components fitted by a mixture model. Supplementary Table S3 and its accompanying text report the exact tolerances, repeated-value frequencies, and odd/even grid pattern. Without the five individual ratings, we cannot infer how that numerical pattern arose.

We compared bounded models by their ability to predict held-out scores, measured by ten-fold expected log predictive density (ELPD); higher values indicate better predictive performance. Every model assigned separate probabilities to exact 0 and exact 1 and represented scores between them with a density on $(0, 1)$. This common treatment made their predictive scores comparable. The candidate set comprised beta and Kumaraswamy densities, a logit-normal density, a logit kernel density, a cross-validated Bernstein-polynomial density, logit-GMMs, and truncated-normal GMMs (TN–GMMs).

The comparison did not resolve a uniquely supported latent-category count. The five-component truncated-normal density ranked highest. Four interior components predicted better than three ($\Delta\text{ELPD} = 8.58$; approximate 95% CI [1.33, 15.82]), but the five-minus-four contrast was not clearly separated from zero ($\Delta\text{ELPD} = 9.51$; approximate 95% CI $[-1.37, 20.38]$). The flexible Bernstein density ranked between the four- and three-component versions and was not clearly separated from the latter (Figure 1; Supplementary Table S3).

The fifth component helps explain why predictive rank does not directly establish another category. It was narrow, centered near 0.5, and reached the fixed 0.005 SD floor in seven of ten folds. Its alignment with the 17 records scored exactly 0.5 is consistent with accommodating that concentration of scores. The models approximate the distribution of the released values; their ranking does not identify the process that generated the ratings. Here, component count refers to the density approximation between 0 and 1, not to the number of descriptive partitions used below. Full model results, uncertainty qualifications, and component diagnostics are reported in Supplementary Table S3.

Separately, under the specified four-Gaussian reference implementation, the in-sample Bayesian information criterion (BIC) was lowest at four components. Bootstrap agreement with the reference assignment had a mean adjusted Rand index (ARI) of 0.917 (2.5th–97.5th percentiles across 500 bootstrap samples: 0.506–0.995). Supplementary Fig. S4 shows both diagnostics, which are conditional on this implementation. For secondary groupwise summaries, we use this reference assignment as an exploratory scaffold. Its three below-ceiling regions were labeled *Idle*, *Routine*, and *Interaction* after fitting. Idle recalls phrases such as "nothing interesting" and "not necessary" in the lowest-score region; Routine reflects recurring "ahead" and "slowing" wording in the next region (243 and 222 of its 494 records, respectively). Interaction is a mnemonic for the upper below-ceiling region, where the lexical profiles include maneuver-related terms such as "merge" and "overtaking." Critical denotes the observed score-1.0 ceiling. These post hoc names aid navigation through the summaries; they do not identify validated cognitive states or semantic classes. Figure 1 and Table 1 summarize this four-partition organization.

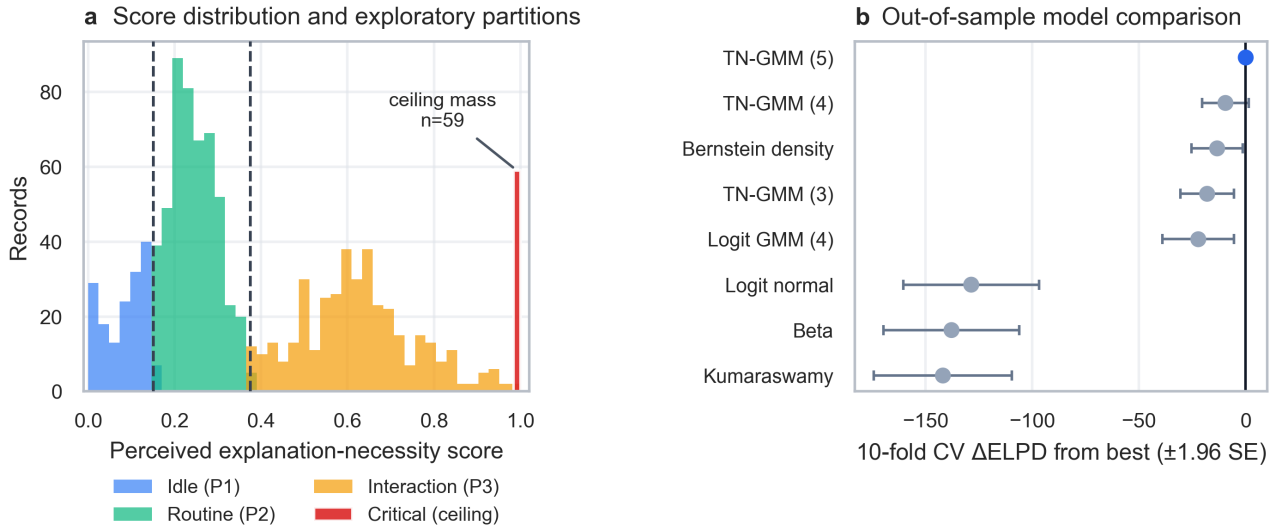


Figure 1. Score distribution and bounded model comparison. (a) Empirical perceived explanation-necessity scores colored by the four-component reference GMM assignment. Dashed lines mark the descriptive Idle–Routine and Routine–Interaction boundaries; the red bar is the observed score-1.0 ceiling mass. (b) Selected ten-fold ELPD differences relative to the best-predicting model, TN–GMM (5); horizontal bars are ±1.96 standard errors of the pointwise held-out difference, conditional on the fixed ten-fold assignment. The colors in panel (a) are a descriptive overlay of the reference assignment, and Critical denotes the observed score-1.0 ceiling category. The full fourteen-model comparison is reported in Supplementary Table S3 and the accompanying text.

Higher necessity scores accompany longer annotations

Explanations were longer in records with higher aggregate necessity scores (Figure 2a). The 1,103 explanations contained 9,541 whitespace-delimited tokens (mean 8.65, range 2–20). Token count correlated positively with the aggregate score (Pearson's $r = 0.374$; template-cluster bootstrap 95% interval [0.266, 0.448]), and the rank correlation gave a similar result (Spearman's $\rho = 0.349$). The bootstrap resamples templates, so records with the same normalized message are drawn together. Record-level Pearson and Spearman p-values and the Fisher-z interval are reported in Supplementary Table S4. We call this association *progressive elaboration*. The term refers only to annotation length, not to the informativeness or quality of what was written.

The partition summaries show where the length differences occurred: Idle and Routine both averaged about 7.7 tokens, whereas annotations in Interaction and Critical were roughly 30% and 45% longer, respectively (Table 1). A descriptive one-way analysis gave $F(3, 1099) = 62.03$, $p = 4.7 \times 10^{-37}$, and $\eta^2 = 0.145$ (template-cluster bootstrap 95% interval [0.074, 0.208]). The standardized Idle–Routine difference was close to zero ($d = 0.005$; template-cluster interval [−0.425, 0.326]),

Partition	n	Weight	Score mean	Score range	Tokens, mean (SD)	Record-weighted TF–IDF distinctiveness
Idle (P1)	163	14.8%	0.083	0.000–0.151	7.71 (2.92)	0.831
Routine (P2)	494	44.8%	0.246	0.152–0.375	7.69 (2.75)	0.774
Interaction (P3)	387	35.1%	0.623	0.380–0.986	9.89 (3.09)	0.928
Critical (ceiling)	59	5.3%	1.000	1.000–1.000	11.15 (2.72)	0.928

Table 1. Descriptive properties of the exploratory score partitions. Explanation length is the count of whitespace-delimited tokens before lexical preprocessing. The displayed term frequency–inverse document frequency (TF–IDF) lexical-distinctiveness estimates are record-weighted, use the harmonized 31-entry conceptual stopword inventory, and equal one minus mean within-partition cosine similarity. The lexical column excludes the sole all-zero TF–IDF vector ($n = 1{,}102$); template-cluster and unique-template sensitivities are reported in Figure 3 and Supplementary Table S6. Critical is the observed ceiling category.

whereas the other five pairwise intervals excluded zero. Supplementary Table S4 reports all six contrasts and both the analytic and template-cluster intervals. These comparisons describe differences within the reference assignment.

The continuous association was not confined to the ceiling category: after those observations were excluded, it remained positive ($n = 1{,}044$, $r = 0.336$, $p = 6.9 \times 10^{-29}$). A regression with standard errors clustered by template also gave a positive standardized slope (0.374; 95% CI [0.273, 0.476]; $p = 5.1 \times 10^{-13}$). Complete sensitivity results are provided in Supplementary Table S12.

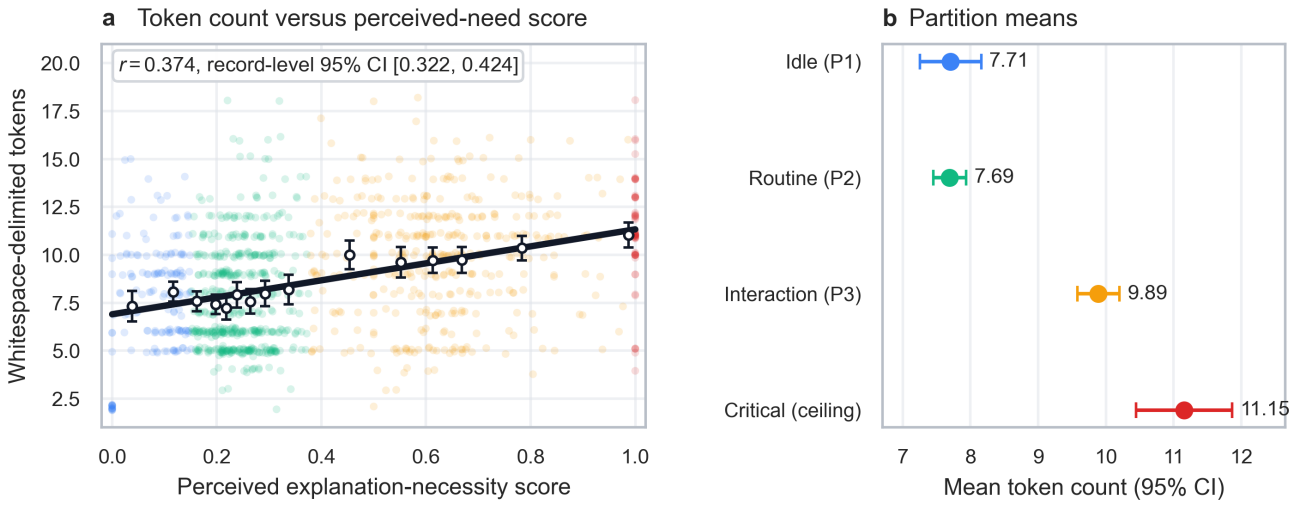


Figure 2. Progressive elaboration in human-written annotations. (a) Whitespace-delimited token count versus perceived explanation-necessity score. Points are lightly jittered vertically for visibility; the line is the ordinary least-squares trend and open points are equal-frequency-bin means with 95% intervals. The annotated interval for Pearson's r is the Fisher-z 95% CI under record-level exchangeability; the template-cluster bootstrap interval [0.266, 0.448] is reported in the text and Supplementary Table S4. (b) Descriptive exploratory-partition means with record-level t-based 95% intervals. Critical is the score-1.0 ceiling category.

Idle–Routine length contrast

The Idle minus Routine mean difference was 0.013 whitespace-delimited tokens (record-level Welch 95% CI [−0.499, 0.525]; template-cluster sensitivity 95% CI [−1.035, 1.062]). No equivalence margin was documented before analysis or independently validated for this domain. Equivalence was not established at the illustrative ±0.25-token bound. In a post hoc check, the record-level 90% interval fell within ±0.5 token, but the wider template-cluster 90% interval did not. We therefore do not infer practical equivalence or a universal explanation-length floor; the complete post hoc margin grid is reported in Supplementary Fig. S1 and Supplementary Table S5.

TF–IDF lexical distinctiveness across exploratory partitions

We represented each explanation as a vector of term frequency–inverse document frequency (TF–IDF) weights[15]. We measured lexical distinctiveness as one minus an explanation's mean cosine similarity to the other explanations in its partition[16]. A higher value means that an explanation shares less weighted vocabulary with the other texts in that partition. The measure concerns lexical overlap, not semantic complexity. One Routine record (`test_1037`) produced the sole all-zero TF–IDF vector, for which cosine similarity is undefined; it was excluded only from the cosine-based lexical analyses, and TF–IDF was refitted on the remaining 1,102 records.

We summarized lexical distinctiveness in two ways. Record weighting counts every explanation, including repeated wording. Unique-template weighting counts each normalized wording once in every partition where it occurs. For example, the largest template appeared in 126 records: 12 Idle, 93 Routine, 20 Interaction, and one Critical. Those frequencies contribute to the record-weighted analysis, whereas the unique-template analysis includes that template once in each of the four partitions. This resembles counting occurrences versus distinct forms, but it is not the conventional token/type distinction: a template is a normalized whole message and may contribute to several partitions. Across the 593 analyzable templates, there were 659 such template–partition memberships.

Both summaries placed Interaction and Critical above Idle and Routine (Figure 3a; Supplementary Table S6). The record-weighted estimates for Idle, Routine, Interaction, and Critical were 0.831, 0.774, 0.928, and 0.928. The unique-template analysis retained the upper–lower pattern but reversed the point-estimate ordering of Idle and Routine. The template-aware pairwise intervals in Supplementary Table S6 excluded zero for all four upper-versus-lower contrasts in both summaries, but not for the Idle–Routine or Interaction–Critical contrasts. Thus, the consistent descriptive finding was upper–lower separation, not a complete four-level ordering.

Because each distinctiveness value depends on the other texts in its partition, we assessed the overall between-partition difference by permutation rather than ordinary ANOVA calibration. The upper partitions also contained longer explanations, so we removed the fitted linear association between distinctiveness and token count before calculating the omnibus statistic. Each reshuffle recomputed the distinctiveness scores, their adjustment for token count, and the omnibus statistic. The record analysis shuffled labels; the template analysis shuffled each template's complete pattern of partition memberships. The resulting plus-one Monte Carlo values were $p = 0.0005$ for records ($F = 114.26$; zero exceedances in 1,999 permutations) and $p = 0.0001$ for templates ($F = 73.78$; zero exceedances in 9,999 permutations).

We also repeated each test with shuffling restricted to equal-length records or templates. For a template, length was its *modal token count*: the most frequent original token count among its records, with ties resolved to the smallest count. Original token counts can differ within a template because punctuation separated by spaces counts as a token before normalization. These restricted shuffles preserved each partition's token-count distribution and gave the same Monte Carlo values. Methods and Supplementary Section 6 describe the procedures and their exchangeability assumptions. The omnibus tests use token-count-adjusted statistics; the estimates and pairwise intervals in Figure 3 and Supplementary Table S6 remain on the original, non-residualized scale. The pairwise intervals describe individual contrasts and are not simultaneous multiplicity-adjusted intervals.

To complement these distribution-level summaries with directly inspectable lexical features, we ranked unigrams for each partition by their one-versus-rest mean TF–IDF ratio (Figure 3b). The resulting high-ratio unigrams describe each partition's lexical profile. A large ratio means that a word has greater mean TF–IDF weight in that partition than in the pooled remaining records, with a small constant added to both means. When a term has little or no weight in the other partitions, the denominator is close to the added constant, 0.001, so a large ratio can arise from a term used in only a few records. This explains the large Idle ratios: for example, *necessary* occurred in only six analyzable records, all in Idle, and always in "not necessary." This between-partition concentration differs from the within-partition overlap measured in panel (a); it does not make Idle the most lexically distinctive partition. Concordance also showed that *interesting* always occurred in "nothing interesting." Complete ratios and document-frequency support for all displayed terms, together with local contexts including the traffic-light uses of *green*, are in Supplementary Tables S8–S9.

Because concordance showed that several high-ratio unigrams occurred in negated phrases, we audited sensitivity to the analyzer's treatment of negators. The negators *no* and *not* were retained in the primary specification. Additionally retaining every analyzer form generated from *don't* changed the distinctiveness estimates by at most 0.00031 and preserved the five highest-ratio unigrams in every partition. The plus-one permutation probability remained $p = 0.0005$, and the full-sample co-occurrence networks were unchanged. Supplementary Tables S7 and S10 and Supplementary Fig. S2 report the complete sensitivity results.

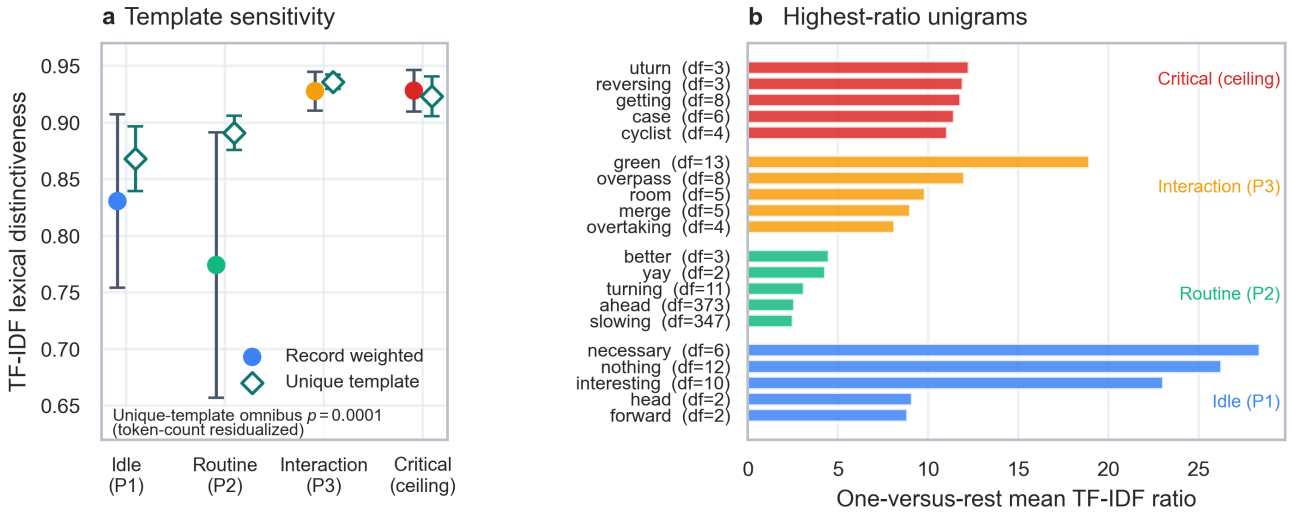


Figure 3. Corpus-specific lexical profiles. The sole all-zero TF–IDF vector was excluded and the vectorizer was refitted on 1,102 records (593 analyzable normalized-message templates). (a) Filled circles are record-weighted TF–IDF lexical-distinctiveness estimates with 95% template-cluster jackknife intervals obtained by deleting all records sharing one global normalized-message template. Open diamonds count each normalized-message template once per partition and use leave-one-global-template-out intervals. Because TF–IDF was fitted separately for the two estimands, absolute levels should not be compared across marker types; interpretation concerns partition contrasts within each estimand and their qualitative agreement. Points and intervals are shown on the original, non-residualized distinctiveness scale. The annotated p-value is the plus-one Monte Carlo probability for the unique-template omnibus statistic after adjustment for token count. It uses 9,999 global permutations, each reassigning complete partition-membership profiles and recalculating all scores and the adjustment. (b) Five highest one-versus-rest mean TF–IDF-ratio unigrams for each exploratory partition. Each ratio compares a term's mean weight in one partition with its mean in all other records pooled together, adding 0.001 to both means. Large Idle ratios reflect concentration relative to that complement, sometimes in very few records; they do not indicate the greatest overall lexical distinctiveness in panel (a). In panel (b), df denotes corpus document frequency across the 1,102 analyzable records; partition-specific document frequencies and local contexts are reported in Supplementary Tables S8–S9.

Leakage-free text prediction across exploratory partitions

We next assessed how well explanation text predicted the reference assignment out of sample. In five-fold cross-validation, records sharing a template stayed together, so no template appeared in both training and held-out data within a fold. All learned text features and classifiers were fitted within the training folds, as detailed in Methods.

The unweighted TF–IDF model achieved balanced accuracy 0.382 (template-cluster bootstrap 95% interval [0.353, 0.403]) and macro-F_1 0.355 [0.323, 0.380] (Table 2). Balanced accuracy averages the proportion correctly classified within each partition, giving the four partitions equal weight. The model's balanced accuracy exceeded the token-count comparator by 0.055 [0.033, 0.083] and the stratified dummy by 0.135 [0.095, 0.168]. Performance was uneven across partitions: the unweighted model identified none of the ceiling records. Class weighting increased ceiling recall from 0.000 to 0.305 and balanced accuracy to 0.450 [0.399, 0.494], but reduced overall accuracy from 0.598 to 0.466. Ceiling precision was low at 0.133 [0.078, 0.200]. These results describe how well the text predicts the exploratory reference assignment, including the tradeoff between overall accuracy and minority-class detection.

The intervals resample templates from predictions already obtained in cross-validation; they do not include variability from refitting the models or redrawing folds. A conventional split by individual records had higher accuracy (0.620), but 53.1% of held-out records then shared a template with training data. We report that specification as a sensitivity rather than as evidence of prediction on unseen wording. Supplementary Table S11 and Fig. S3 provide all classwise results, comparator intervals, and the secondary permutation test.

Out-of-fold model	Balanced accuracy	Macro-F_1	Overall accuracy	Ceiling precision	Ceiling recall
TF–IDF logistic	0.382	0.355	0.598	0.000	0.000
TF–IDF logistic, class weighted	0.450	0.406	0.466	0.133	0.305
Token-count logistic	0.327	0.291	0.529	0.000	0.000
Training-fold most frequent	0.190	0.166	0.316	0.000	0.000
Stratified dummy	0.247	0.246	0.327	0.074	0.085

Table 2. Template-grouped, leakage-free classification performance. Point estimates use locked five-fold out-of-fold predictions, with no normalized-message template appearing in both the training and held-out sets within any cross-validation fold. The training-fold most-frequent baseline predicts the majority class observed within each training fold; because template grouping changes fold-specific class composition, its pooled balanced accuracy need not equal 0.25. The Ceiling columns report precision and recall for Critical, the score-1.0 ceiling category. Balanced accuracy and macro-F_1 are reported before overall accuracy because this category contains only 59 observations. Template-cluster bootstrap intervals, all classwise recalls, and ceiling-category precision are in Supplementary Table S11; complete per-class precision/recall and record-bootstrap results are in the verification archive.

Release-complete cohort sensitivity

Applying the same pipeline to all 1,280 released records preserved the main findings. The score–length correlation remained positive ($r = 0.382$; template-cluster bootstrap 95% interval [0.272, 0.453]), including without the ceiling observations ($r = 0.338$). Both lexical summaries again showed upper–lower separation, and all four permutation procedures gave zero exceedances. The grouped text models again exceeded the token-count and dummy comparators in balanced accuracy and macro-F_1.

The two cohorts differed in which adjacent component counts were clearly separated in held-out prediction. In the complete release, the TN–GMM five-minus-four contrast was clearly separated from zero, whereas the four-minus-three contrast was not. The fifth component again aligned with the exact-0.5 concentration and reached the SD floor in all ten folds. These results still left the latent-category count unresolved. Supplementary Section 14 and Table S14 report the cohort-specific estimates and contrasts side by side.

Discussion

The primary empirical result is the association between aggregate perceived explanation necessity and the length of human-written annotations. This association makes annotation length worth considering when selecting candidate messages for listener studies. Those studies must determine which details to retain and how much text to deliver. The association appeared in both Pearson and rank correlations, remained after excluding the ceiling category, and remained positive when repeated wording was treated as clustered observations. The result therefore concerns the continuous score–length relationship, independently of the descriptive partitions. It also persisted in the complete release. *Progressive elaboration* denotes the observed record-level association between aggregate perceived explanation-necessity scores and explanation length; its implications for individual writing behavior and explanation utility require separate study.

Because the score–length association does not depend on a categorical interpretation, the evidential status of the reference partitions can be considered separately. The specified reference implementation provides a traceable four-partition organization: three reference GMM regions lie below the ceiling, and the fourth is the observed maximum-score category. Computational reproducibility within a chosen model family does not by itself validate a categorical interpretation. The bounded comparison illustrates why this distinction matters. Extra mixture components can accommodate the shape and repeated values of a score distribution, consistent with the narrow midpoint component here, rather than necessarily identify another substantive category[12]. Moreover, the two cohorts differed in which adjacent component counts were clearly separated in held-out prediction. The reference partitions therefore serve to organize secondary summaries, while the number of latent categories remains unresolved.

Across the two lexical analyses, the common descriptive result was upper–lower separation rather than a complete four-level ordering. The two summaries offer complementary views of lexical overlap. The record-weighted summary reflects overlap among all observed messages, including repetitions; the unique-template summary reflects overlap among distinct normalized wordings within each partition. Their agreement on upper–lower separation shows that this descriptive pattern persists when repeated messages no longer contribute in proportion to their frequency. The reversal of the Idle–Routine point estimates shows that their relative lexical distinctiveness depends on how repeated wording contributes to the summary. Neither approach separated Interaction from Critical.

The predictive analysis asks a complementary question: can wording from templates unseen during training predict the reference assignments? Performance above the declared comparators indicates corpus-specific lexical regularities, while the low ceiling precision and the class-weighting tradeoff limit its use as a classifier of rare ceiling records. Together, the lexical summaries and prediction results help characterize candidate text for further study; they do not evaluate its usefulness to a passenger. Prior work on mechanistic, teleological, and counterfactual explanations motivates future coding and experimental comparison, not a reinterpretation of these unigrams[17].

Retrospective annotation is useful here because it elicits candidate explanatory content across recorded scenes. The present corpus provides a record-level view of how the length of that elicited text covaries with aggregate necessity judgments, together with a secondary description of its lexical distribution. Using annotation-derived content for recipient-facing design, however, requires prospective validation. Experimental studies of delivered explanations show that effects depend on content, perceived risk, timing, and modality[2,4–6]. That validation must account for different cognitive demands: a passenger as listener may process an explanation while monitoring a dynamic environment, whereas an annotator as speaker–observer writes after viewing a recorded scene without that concurrent performance demand. The two perspectives can therefore be linked in a recipient-facing research framework: annotation-derived text supplies candidate content, whereas prospective listener studies determine how that content should be selected, compressed, prioritized, timed, or staged during travel.

Such a study could draw candidate content from records across necessity levels, independently vary its delivery in a simulator or controlled on-road study, and measure comprehension, workload, calibrated reliance, monitoring behavior, and driving or takeover performance. This design would test which annotation-derived details should be preserved for listeners without assuming that longer retrospective annotations should become longer delivered messages. Trust outcomes should operationalize calibration as appropriate reliance aligned with demonstrated system capability rather than as uniformly higher self-reported trust[18].

Limitations. Several limitations should be noted. First, the missing writer identifiers and individual ratings prevent models of repeated observations from the same annotator. The documented five-person pool also limits generalization beyond the source annotators. The score–length result remains a corpus-level association, and template clustering addresses only observable wording reuse. Because the release does not show how one explanation was retained from the candidates, that selection may partly account for the observed association. Both cohort scopes preserved the main conclusions, but their estimates and adjacent bounded-model contrasts are not interchangeable. The available source documentation also does not state whether annotators received target-length or conciseness instructions; any such instruction could have constrained the observed length distribution.

Second, the four descriptive partitions arise from an exploratory approximation to one bounded scalar and should not be read as validated psychological categories. The post hoc labels summarize three below-ceiling score regions and the score-1.0 ceiling category; the latter's small size limits comparisons and classification. Accordingly, the lexical contrasts and classification performance are conditional on this exploratory reference assignment; partition-independent text outcomes would constitute different estimands and are not evaluated here.

Third, this is a retrospective annotation corpus. It does not measure objective hazard, cognitive state, listener comprehension, trust calibration, or real-time driving performance. Causal design recommendations require prospective experiments.

Fourth, the text analyses have distinct measurement and inferential limits. At the measurement level, whitespace-delimited token count directly measures text length but not information quality. TF–IDF distinctiveness measures weighted lexical overlap, whereas co-occurrence density measures the proportion of possible pairs among the 12 selected top tokens that co-occurred in at least two explanations; neither measures semantics, syntax, or communicative effectiveness. Unigram ratios are context-sensitive and corpus-specific. For inference, the global lexical permutations assumed exchangeability across records or templates and did not preserve each partition's token-count distribution. The additional tests restricted shuffling to equal token counts, preserved those distributions, and yielded the same plus-one Monte Carlo values.

Fifth, no equivalence margin was documented before analysis or validated for this domain. The margin grid is retained only as a post hoc sensitivity: equivalence was not established at ± 0.25 token, and the record-level 90% interval fell within ± 0.5 token, whereas the template-cluster 90% interval did not.

Finally, the data are short English-language explanations from one public collection. Lexical results may depend on language, corpus, repeated templates, and preprocessing. Record weighting treats repeated wording as part of the empirical explanation distribution, whereas unique-template weighting describes an observed lexical repertoire. Both estimands and the preprocessing specifications used in the reported analyses are supplied for reproducibility.

Conclusion

In the 1,103-record primary cohort, higher aggregate perceived explanation-necessity scores were associated with longer human-written retrospective explanations; the association remained positive under template clustering and in the analysis of all 1,280 released records. Bounded out-of-sample comparisons did not resolve a unique latent-category count. For groupwise summaries, we used a descriptive four-partition scaffold comprising three below-ceiling regions and the score-1.0 ceiling

category. Both lexical analyses showed upper–lower separation rather than a complete four-level ordering. Our audit describes the documented annotation protocol, how often wording recurs, and how the scores are quantized. The executable workflow for both cohorts provides a reproducible reference for future analyses of this public corpus. These corpus-level patterns motivate prospective studies of how annotation-derived content should be selected, compressed, and delivered under real-time listener constraints.

Methods

Dataset, annotation provenance, and record audit

We analyzed the public `explanation_necessity_labels.txt` file linked by the source project, an explanation-annotation layer over Berkeley DeepDrive Attention (BDD-A) videos[8,19]. The public file contained 1,280 line-oriented Python dictionaries: 827 training, 276 test, and 177 validation records. The source article documents a 1,103-clip corpus[8]; the current release contains 1,280 records, and its `training_` and `test_` prefixes identify a 1,103-record subset. The available documentation does not provide a row-level mapping between the source-described corpus and the current release splits. We used the release-defined training-plus-test subset as the operational primary cohort; its boundary depended only on split metadata, not scores, message content, or fitted-model results. All 1,280 released records were analyzed under the same pipeline as a cohort-scope sensitivity. For the primary analysis, we parsed each selected line with a literal parser, required a non-empty message, a score in $[0, 1]$, and a unique `vid`, and retained all 1,103 selected records. Analyzed-record tables, selection rules, source and cohort checksums, split counts, and the acquisition URL are supplied in the verification archive.

The source documentation reports five college-student annotators, aged 18–25, who held U.S. driver's licenses. The first archived version states that each annotator completed all 1,103 clips, implying 1,103 raw annotation tasks per annotator and 5,515 clip–person tasks in total[8]. In each task, an annotator selected a moment, entered an explanation-necessity score, and wrote a first-person explanation. The five scores were aggregated by removing the largest and smallest and averaging the remaining three[8]. The public release exposes only `vid`, `timestampInVideo`, `message`, `video_url`, and `critical_score`; it does not expose annotator identifiers, the five individual ratings, the candidate explanations, or the linkage identifying which explanation was retained. Consequently, we cannot recover how many retained explanations each writer contributed or link a retained explanation to its writer's own necessity rating.

A normalized-message template groups messages that become identical after a fixed text-cleaning rule. We lowercased messages and deleted straight and curly single and double quotation marks (including apostrophes), commas, exclamation marks, periods, ASCII hyphen-minus characters, and question marks. We then collapsed whitespace by splitting and rejoining with single spaces; no other punctuation was removed. Each distinct cleaned message defined one template, and records sharing it formed a template cluster. The 594 templates supplied the grouping units for score–length sensitivity and grouped cross-validation. After the cosine-only zero-vector exclusion, the 593 remaining templates supplied the grouping and equal-template units for lexical analysis. Template-based grouping addresses observable wording reuse only.

Token count and preprocessing policy

Explanation length was defined before stopword or punctuation processing as `len(message.split())`: the number of whitespace-delimited tokens.

Lexical analyses used one 31-entry conceptual stopword inventory, reported as an implementation registry rather than as linguistically homogeneous stopword classes:

> the, a, an, is, are, will, be, have, has, to, for, of, in, on, at, and, or, so, i, me, my, we, our, it, that, this, there, because, now, let, don't.

The inventory was compiled into analyzer-specific configured strings. For the primary TF–IDF path, we lowercased text and deleted straight and curly double quotation marks, commas, exclamation marks, periods, ASCII hyphen-minus characters, and question marks while preserving both straight and curly apostrophes; scikit-learn's default token analyzer then treats either apostrophe as a token boundary (for example, the ASCII-apostrophe form `I'll` and the curly-apostrophe form `I’ll` both emit `ll`). The configured 37-string profile is a surface compilation of this single conceptual policy, including possible analyzer fragments and merged spellings. For the co-occurrence analysis, the token analyzer instead deletes straight and curly apostrophes and can merge *don't* into *dont*. In the primary compiled profiles, the standalone negators *no* and *not* were retained, whereas *don*, *don't*, and *dont* were filtered as analyzer surface forms derived from the registry entry *don't*. We therefore repeated the lexical-distinctiveness and co-occurrence analyses while additionally retaining *don*, *don't*, and *dont*, because negation can reverse proposition polarity. The exact configured profiles used in the reported analyses are in Supplementary Table S2 and the code archive.

Exploratory GMM partitions

To reproduce the reference partition, we fitted a one-dimensional four-component `GaussianMixture` to all 1,103 scores. The settings were `covariance_type='full'`, `init_params='kmeans'`, `n_init=10`, `random_state=42`, and `reg_covar`=10^{-6}. Hard assignments used the largest posterior responsibility, and components were ordered by mean assigned score. Only after fitting was the highest component—which comprised exactly the 59 score-1.0 records—designated the Critical ceiling category. Standard in-sample BIC was computed for one through eight components using the same settings; 500 nonparametric bootstrap samples assessed agreement with the four-component reference using ARI. These diagnostics are conditional on the specified reference implementation and are not evidence of latent categories.

The partition names were assigned after fitting to summarize score position and recurring wording. Their rationale is given when the partitions are introduced in Results; supporting frequencies and lexical contexts are reported in Supplementary Tables S8–S9.

Bounded out-of-sample model comparison

To characterize the finite precision of the released scores before comparing bounded candidates, we projected each value to its nearest k/d point for candidate integer denominators $d = 2,\ldots,5{,}000$ and applied an absolute tolerance of 10^{-7}. Denominator 1,836 was the smallest for which 1,102 values met the tolerance; the sole exception was the released value 0.3. This is a descriptive finite-precision audit, not a reconstruction of the annotation interface or raw-score aggregation.

We compared predictive performance by cross-validation[20], using ten fixed folds stratified by boundary status and eight interior score quantiles (seed 20260815). Every candidate used probability masses at exactly 0 and 1 plus a normalized density on $(0,1)$, placing all models on the same mixed reference measure. Interior candidates were beta, Kumaraswamy[21], logit-normal, logit kernel density, cross-validated Bernstein-polynomial beta-basis density[22], logit-GMMs with two to five components, and truncated-normal GMMs with one to five components. Within each fixed outer training fold, a 0.5-count Jeffreys correction estimated the probabilities of exact 0, interior, and exact 1, and all interior-density parameters were fitted only on training-interior values. Five-fold inner cross-validation selected the kernel bandwidth and Bernstein degree. Records held out in the outer folds were used only for log scoring. Truncated-normal component SDs were bounded below by 0.005 to prevent a repeated exact score from producing a degenerate likelihood. The resulting TN–GMM comparisons describe predictive performance under this specified lower bound. We summed held-out log predictive density. For paired model comparisons, with δ_i denoting the held-out log-score difference for record i, we computed $\mathrm{SE}(\Delta\mathrm{ELPD}) = \sqrt{N s_\delta^2}$, where $N = 1{,}103$ and s_δ^2 is the sample variance of the pointwise differences. These pointwise standard errors are approximate and conditional on the single fixed ten-fold assignment; they do not incorporate variability across alternative fold partitions or overlapping training fits. Because the released interior score is numerically quantized, these candidates are comparative continuous-density approximations on their common reference measure rather than reconstructions of the acquisition mechanism. A raw unconstrained Gaussian score was retained only as a non-comparable sensitivity because its endpoint density depends on variance regularization and measurement units.

Length statistics, effect intervals, and dependence sensitivity

We calculated Pearson and Spearman correlations between score and token count. We used record-level t-based 95% intervals for reference-partition means; η^2 has a noncentral-F interval; standardized mean-difference intervals invert the noncentral t distribution. Kruskal–Wallis and negative-binomial models are supplied as distributional sensitivities. For these tests, we report computed p-values rather than threshold statements.

For visible text reuse, we fitted the score–token-count association with template-cluster-robust standard errors and resampled the 594 normalized-message templates with replacement 10,000 times (seed 20260815). Whenever a template was selected, all of its records were included together across every partition in which it occurred. For each replicate we recomputed the correlation, record-weighted four-partition η^2, and pooled-SD Cohen's d for all six contrasts, conditional on the fixed reference assignments, and report 95% percentile intervals. This sensitivity addresses observable within-template replication but does not assess latent-category validity.

Post hoc margin sensitivity

We estimated the Idle-minus-Routine mean token-count difference using Welch's method and report 95% and 90% confidence intervals[23,24]. Because no independently established practical margin was available and no margin was documented before analysis, we did not treat the two one-sided tests (TOST) procedure as a confirmatory test of practical equivalence. As a descriptive sensitivity, we compared the record-level 90% interval with illustrative post hoc bounds of ±0.25, ±0.5, ±0.75, and ±1.0 token and report the corresponding TOST p-values. We also report a template-cluster-robust interval as a sensitivity to observable wording reuse.

TF–IDF lexical distinctiveness and full-recalculation permutations

We initially fitted a TF–IDF vectorizer[15] to all harmonized primary documents (minimum document frequency 2; maximum 500 features). The sole all-zero vector, `test_1037`, makes cosine similarity mathematically undefined; we therefore excluded that record only from cosine-based lexical analyses and refitted the vectorizer on the remaining 1,102 documents. The record remained in the provenance, token-length, and classification analyses. For record i in a partition of size n_g, lexical distinctiveness was

$$d_i = 1 - \frac{1}{n_g - 1} \sum_{j \neq i,\, g_j = g_i} \cos(\mathbf{v}_i, \mathbf{v}_j). \tag{1}$$

The partition estimate is equivalently one minus the mean off-diagonal cosine similarity, a pairwise U-statistic. For jackknife uncertainty calculations, the relevant fitted TF–IDF representation and fixed reference assignments were held fixed; the vectorizer was not refitted. Record-level intervals used partition-specific leave-one-record U-statistic replicates. Template-cluster intervals deleted all records sharing one normalized-message template and recomputed the affected partition means and between-partition contrasts. Figure 3 uses the latter intervals. To adjust the omnibus comparison for token count, we regressed d_i on token count and computed an omnibus F statistic from the residuals.

Because each d_i is calculated relative to other records, the nominal residual-ANOVA reference distribution does not represent this dependence. We instead permuted partition labels across records 1,999 times, preserving overall partition sizes and keeping each text paired with its observed token count. After every shuffle, we recomputed all d_i values, refitted the token-count regression, and recalculated F. The plus-one Monte Carlo probability was $(b+1)/(B+1)$, where b is the number of permuted statistics at least as large as observed[25]. This global permutation incorporates the dependence created by group-relative scoring. It assumes that record labels are exchangeable across the cohort; it does not preserve, or condition on, each partition's observed token-count distribution.

The record-weighted estimand counts every analyzable explanation in the analyzed corpus, so recurring messages contribute at their observed within-partition frequencies. We therefore evaluated a complementary unique-template estimand at the global-template level. Among the 593 normalized-message templates analyzable by cosine methods, 54 occurred in more than one partition, so assigning one representative label would discard observed memberships. Each template instead contributed once to every partition in which it occurred, producing 659 template–partition cells (Idle 93, Routine 192, Interaction 316, Critical 58). Because all records sharing a template emitted the same harmonized analyzer-token sequence, we refitted TF–IDF to that shared sequence once per template. Residualization used each template's modal token count: the most frequent whitespace-delimited token count among its original messages, with ties resolved to the smallest count. With the fitted representation and observed partition memberships held fixed, the unique-template jackknife deleted one global template from all partitions simultaneously. We then recomputed the affected partition means and between-partition contrasts without refitting the vectorizer.

For permutation calibration, each template was represented by its complete presence profile, indicating which of the four partitions contained it. We reassigned these profiles jointly among template vectors 9,999 times (seed 20260818), keeping each vector paired with its modal token count. After each reassignment, we recomputed all within-partition scores and repeated the token-count residualization. The shuffles preserved each partition's unique-template count and the pattern of sharing across partitions, but not each partition's modal-token-count distribution. The resulting Monte Carlo probability assumes that complete template-presence profiles are exchangeable across templates; it does not condition on the observed modal-token-count distributions within partitions.

To condition on token count, we repeated both permutation procedures within exact token-count strata (seed 20260825). Record labels were reassigned only among records with the same observed whitespace-delimited token count. Complete four-partition presence profiles were reassigned only among templates with the same modal token count. Thus, each shuffle preserved every partition's full distribution of observed token counts in the record analysis, or modal token counts in the template analysis. After each reassignment, we recomputed all group-relative distinctiveness values, repeated the token-count residualization, and recalculated the omnibus statistic.

Separately from the distinctiveness estimands and their permutation calibration, we conducted descriptive unigram, concordance, and co-occurrence analyses. For each unigram, we calculated its mean TF–IDF weight among records in one partition and among all records in the other partitions pooled together. After adding $\varepsilon = 0.001$ to both means, we divided the first by the second. These one-versus-rest ratios describe how strongly individual unigrams are concentrated in a partition relative to its complement; they are not out-of-sample predictive or causal measures. We audited every occurrence of selected context-sensitive and high-ratio tokens and tabulated adjacent bigrams and centered trigrams. Co-occurrence networks used the 12 most frequent non-stopword tokens as nodes; an edge joined tokens observed in the same explanation and required at least two explanations. Network-density sample-size sensitivity used 500 within-partition bootstrap resamples of size 59, drawn with replacement (seed 42); networks with and without the *don't*-derived forms were evaluated on the same draws. Within each bootstrap draw and network specification, the 12 most frequent non-stopword tokens were re-identified from the sampled

records before density was recomputed. Graph density is the proportion of possible undirected edges that were observed[26]; here it describes top-token overlap and not semantic topology.

Leakage-free text classification

To assess corpus-specific prediction for normalized-message templates unseen during training, we used the fixed four-partition reference assignment described above as the classification target. Predictive performance was interpreted as separation of this exploratory scaffold, not as independent evidence for latent categories. The primary analysis used a fixed StratifiedGroupKFold with five folds, seed 42, and the 594 normalized-message templates as groups. No template appeared in both the training and held-out sets within any fold.

The classifier used a fixed preprocessing rule that did not depend on partition labels. It lowercased messages and removed the non-apostrophe punctuation specified above, but preserved straight and curly apostrophes, unlike the template normalizer. Within each training fold, a Pipeline applied the fixed harmonized stopword surface list and fitted the TF–IDF vocabulary (minimum document frequency 2, maximum 500 features), IDF weights, and multinomial logistic regression. No vocabulary, feature statistic, label-dependent decision, or hyperparameter tuning occurred before splitting[27]. We compared unweighted and inverse-frequency-weighted text classifiers with token-count-only logistic regression, most-frequent prediction, and a stratified dummy. Both text classifiers used multinomial logistic regression with the L-BFGS solver, default L_2 regularization ($C = 1$), a 2,000-iteration limit, and seed 42; the weighted model used inverse-frequency class weights. The token-count comparator learned a standardization of whitespace-delimited token count within each training fold before fitting the same unweighted logistic specification. We report accuracy, balanced accuracy, macro-F_1, and per-class confusion matrices. Template grouping tests generalization to unseen wording templates.

The five fold scores share overlapping training sets and were not treated as independent replicates[28]. We estimated uncertainty by both 10,000 stratified record bootstraps and 10,000 template-cluster bootstraps of locked grouped out-of-fold predictions. These bootstraps resampled observed labels together with their already obtained predictions; they did not refit the five cross-validation pipelines or redraw folds, so the intervals are conditional on the fitted grouped predictions and fixed fold assignment. We also report a record-stratified pipeline as a sensitivity analysis: every learned transformation remained training-fold-only, but 586 records (53.1%) had the same normalized-message template in their training fold. For that secondary specification, a 499-iteration label permutation refitted the classifier for each permutation and tested accuracy improvement over the majority baseline; the fold-specific vocabulary and IDF transformations were cached because they depend only on documents, not permuted labels.

Software and reproducibility

Primary reproduction used Python 3.10.12, NumPy 1.26.4, pandas 2.2.0, SciPy 1.12.0, scikit-learn 1.4.0, statsmodels 0.14.1, Matplotlib 3.8.3, seaborn 0.13.2, NetworkX 3.2.1, and CuPy 13.6.0 in a Docker image. Supplementary Table S13 summarizes the random seeds and iteration counts for each procedure; complete settings are recorded in the verification archive. The same software, settings, and procedure-specific seeds were used for the release-complete rerun. The dual-cohort verification archive contains the exact record identifiers, derived tables, preprocessing and analysis scripts, figure-generation code, environment specification, seeds, source and cohort SHA–256 checksums, and concise validation summaries for both run trees.

Ethics Statement

The study was limited to the secondary analysis of a publicly available label file. The present study involved no recruitment, interaction, intervention, or collection of new data from human participants, and the file contains no annotator identifiers or raw individual ratings. As the public, anonymized label file contains no personally identifiable information, this study is exempt from review by the Institutional Review Board (IRB) of Gwangju Institute of Science and Technology (GIST). No participants were recruited or contacted for this secondary analysis; informed consent for the present analysis was therefore not applicable. All methods were carried out in accordance with relevant guidelines and regulations.

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Author contributions statement

G.K. conceived the study, developed the code, conducted the analysis, and wrote the original manuscript. D.Y., Y.K., and M.O. contributed to conceptual discussion and manuscript review. S.K. supervised the project and edited the manuscript. All authors reviewed and approved the manuscript.

Data availability

The source object is the project's public `explanation_necessity_labels.txt` file (`https://drive.google.com/file/d/1dTUkcyRtwH5wdG8aaz_ISXjaQ-R3yNT1/view`; file modification timestamp 11 November 2022)[8]. The primary analytic dataset is the deterministic 1,103-record cohort selected from that 1,280-row object by `vid` prefix (`training_` or `test_`); the release-complete sensitivity uses all 1,280 rows. The verification archive available under the Zenodo DOI below identifies the acquisition URL, byte counts and SHA–256 checksums, analyzed record identifiers, split membership, and derived data for both cohorts. The archive does not redistribute annotation text; its verifier confirms the source-file checksum and reconstructs both analytic inputs by their declared selection rules.

Code availability

The Dockerized dual-cohort analysis and verification archive available under the Zenodo DOI `https://doi.org/10.5281/zenodo.22105849` contains preprocessing, statistical analysis, sensitivity analysis, figure-generation code, software versions, random seeds, and machine-readable results for the 1,103-record primary cohort and the 1,280-record release-complete sensitivity. Supplementary Table S13 summarizes the deposited materials and execution settings for the analyses reported in the article and Supplementary Information. The archive's top-level `PUBLIC_FILE_MANIFEST.json` and `SHA256SUMS.txt` provide the deposited content-file inventory and SHA–256 identities. The side-by-side robustness results are reported in Supplementary Section 14 and Table S14.

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Competing interests

The authors declare no competing interests.